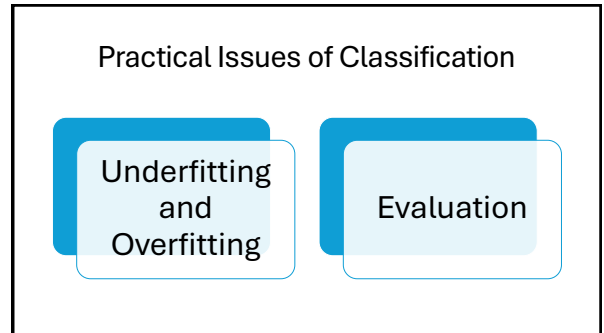
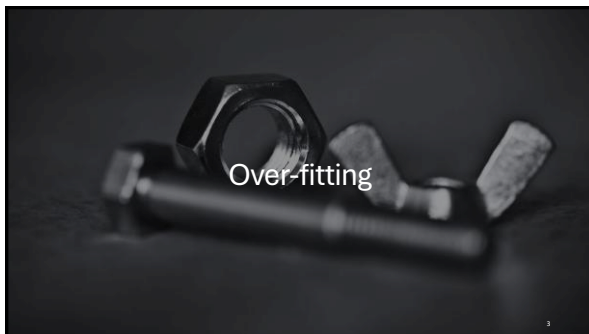




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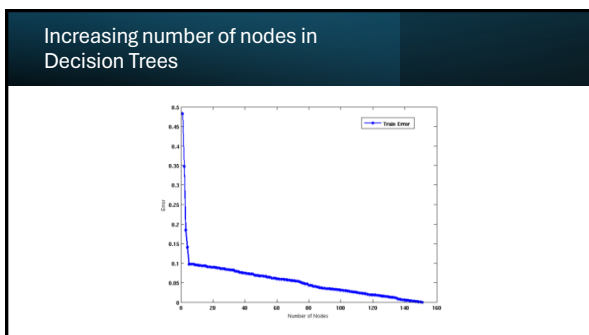


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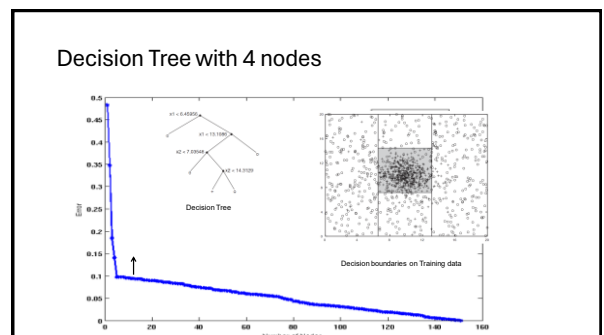
Classification Errors

- Training errors (apparent errors)
 - Errors committed on the training set
- Test errors
 - Errors committed on the test set
- Generalization errors
 - Expected error of a model over random selection of records from same distribution

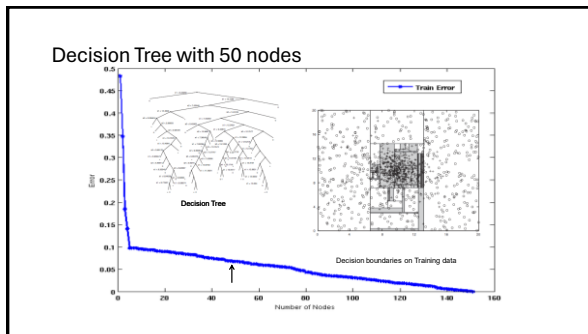
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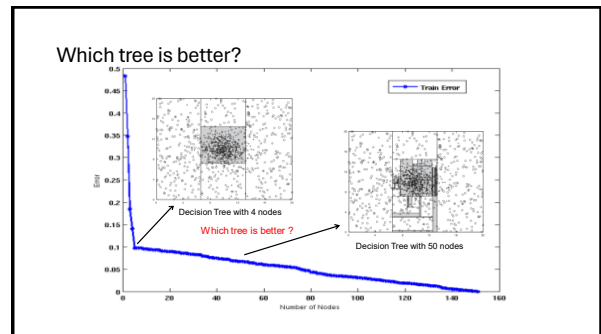
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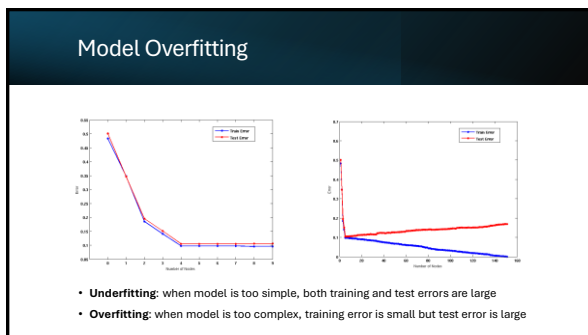
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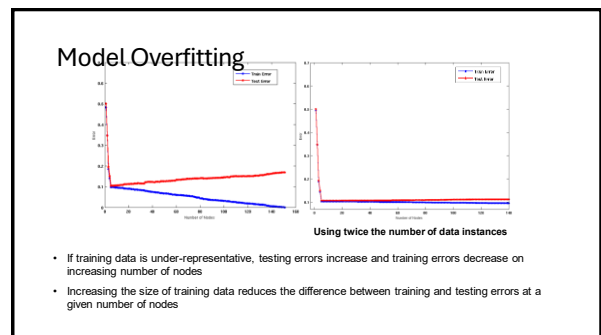
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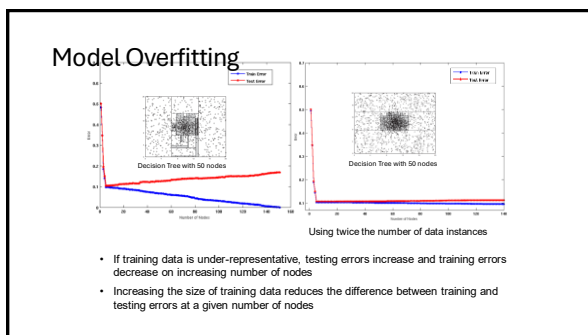
8



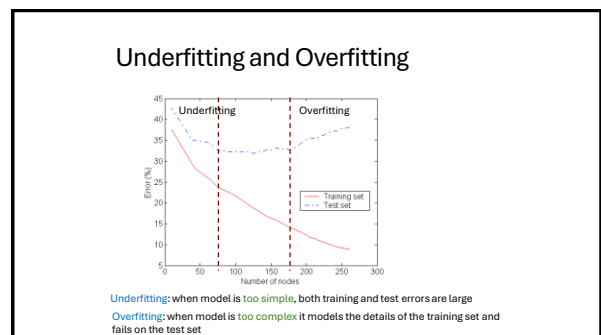
9



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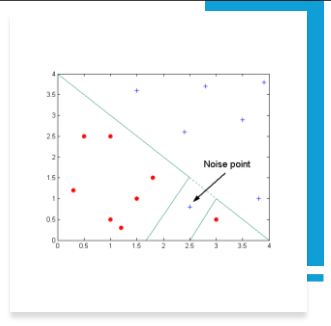
11



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Overfitting due to Noise

Decision boundary is distorted by noise point



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Notes on Overfitting

- Overfitting results in decision trees that are more complex than necessary
- Training error no longer provides a good estimate of how well the tree will perform on previously unseen records
 - The model does not generalize well
- Need new ways for estimating generalization errors
- Reasons
 - Limited Training Size
 - High Model Complexity
 - Multiple Comparison Procedure

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How to Address Overfitting

Prune the Tree OR Prune the Rule

- In order to Reduce the complexity of decision procedure we have two options
 - (i) either we can prune the tree first and then develop the rules or
 - (ii) we can develop the rules and then prune the rules.
- Which is better?
 - (ii) is better, why?

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How to Address Overfitting: Tree Pruning

- **Pre-Pruning (Early Stopping Rule)**
 - Stop the algorithm before it becomes a fully-grown tree
 - Typical stopping conditions for a node:
 - Stop if all instances belong to the same class
 - Stop if all the attribute values are the same
 - More **restrictive** conditions:
 - Stop if **number of instances** is less than some user-specified threshold
 - Stop if class distribution of instances are **independent** of the available features (e.g., using χ^2 test)
 - Stop if expanding the current node **does not improve impurity** measures (e.g., Gini or information gain) or it falls below a threshold value.
- Upon halting, the node becomes a leaf
- The leaf may hold the most frequent class among the subset tuples.
- Problem
 - Difficult to choose an appropriate threshold

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Model Selection for Decision Trees

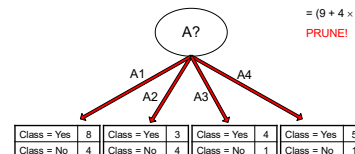
- **Post-pruning**
 - Grow decision tree to its entirety
 - Subtree replacement
 - Trim the nodes of the decision tree in a bottom-up fashion
 - If generalization error improves after trimming, replace sub-tree by a leaf node
 - Class label of leaf node is determined from majority class of instances in the sub-tree
 - Subtree raising
 - Replace subtree with most frequently used branch

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Example of Post-Pruning

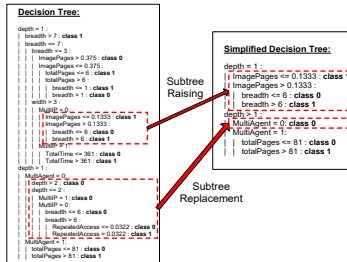
Class = Yes	20
Class = No	10
Error = 10/30	

Training Error (Before splitting) = 10/30
 Pessimistic error = $(10 + 0.5)/30 = 10.5/30$
 Training Error (After splitting) = 9/30
 Pessimistic error (After splitting)
 $= (9 + 4 \times 0.5)/30 = 11/30$
PRUNE!



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Examples of Post-pruning



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Model Evaluation

- Purpose:
 - To estimate performance of classifier on previously unseen data (test set)
- Holdout
 - Reserve k% for training and (100-k)% for testing
 - Random subsampling: repeated holdout
- Cross validation
 - Partition data into k disjoint subsets
 - k-fold: train on k-1 partitions, test on the remaining one
 - Leave-one-out: k=n

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How to Address Overfitting...

- **Post-pruning**
 - Grow decision tree to its entirety
 - Trim the nodes of the decision tree in a **bottom-up** fashion
 - If generalization error improves after trimming, replace sub-tree by a leaf node.
 - Class label of leaf node is determined from majority class of instances in the sub-tree

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Model Evaluation

- Metrics for Performance Evaluation
 - How to evaluate the performance of a model?
- Methods for Performance Evaluation
 - How to obtain reliable estimates?

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Model Evaluation

- **Metrics for Performance Evaluation**
 - How to evaluate the performance of a model?
- **Methods for Performance Evaluation**
 - How to obtain reliable estimates?

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Metrics for Performance Evaluation

- Focus on the **predictive capability** of a model
 - Rather than how fast it takes to classify or build models, scalability, etc.
- **Confusion Matrix:**

	PREDICTED CLASS	
	Class=Yes	Class=No
	Class=Yes	Class=No
ACTUAL CLASS	a	b
	c	d

a: TP (true positive)
 b: FN (false negative)
 c: FP (false positive)
 d: TN (true negative)

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Metrics for Performance Evaluation...

ACTUAL CLASS	PREDICTED CLASS	
	Class=Yes	Class=No
	Class=Yes	Class=No
Class=Yes	a (TP)	b (FN)
Class=No	c (FP)	d (TN)

- Most widely-used metric:

$$\text{Accuracy} = \frac{a + d}{a + b + c + d} = \frac{TP + TN}{TP + TN + FP + FN}$$

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Limitation of Accuracy

- Consider a 2-class problem
 - Number of Class 0 examples = 9990
 - Number of Class 1 examples = 10
- If model predicts everything to be class 0, accuracy is $9990/10000 = 99.9\%$
 - Accuracy is misleading because model does not detect any class 1 example

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Cost Matrix

	PREDICTED CLASS		
ACTUAL CLASS	C(i j)	Class=Yes	Class=No
	Class=Yes	C(Yes Yes)	C(No Yes)
	Class=No	C(Yes No)	C(No No)

$C(i|j)$: Cost of classifying class j example as class i

$$\text{Weighted Accuracy} = \frac{w_a a + w_d d}{w_a a + w_b b + w_c c + w_d d}$$

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Computing Cost of Classification

Cost Matrix	PREDICTED CLASS		
ACTUAL CLASS	C(i j)	+	-
	+	-1	100
	-	1	0

Model M_1	PREDICTED CLASS		
ACTUAL CLASS		+	-
	+	150	40
	-	60	250

Accuracy = 80%
Cost = 3910

Model M_2	PREDICTED CLASS		
ACTUAL CLASS		+	-
	+	250	45
	-	5	200

Accuracy = 90%
Cost = 4255

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Cost vs Accuracy

Count	PREDICTED CLASS	
	Class=Yes	Class=No
	Class=Yes	Class=No
Class=Yes	a	b
Class=No	c	d

Cost	PREDICTED CLASS	
	Class=Yes	Class=No
	Class=Yes	Class=No
Class=Yes	p	q
Class=No	q	p

Accuracy is proportional to cost if

- $C(\text{Yes}|\text{No}) = C(\text{No}|\text{Yes}) = q$
- $C(\text{Yes}|\text{Yes}) = C(\text{No}|\text{No}) = p$

$$N = a + b + c + d$$

$$\text{Accuracy} = (a + d)/N$$

$$\begin{aligned} \text{Cost} &= p(a + d) + q(b + c) \\ &= p(a + d) + q(N - a - d) \\ &= qN - (q - p)(a + d) \\ &= N[q - (q - p) \times \text{Accuracy}] \end{aligned}$$

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Precision-Recall

$$\text{Precision (p)} = \frac{a}{a + c} = \frac{TP}{TP + FP}$$

$$\text{Recall (r)} = \frac{a}{a + b} = \frac{TP}{TP + FN}$$

$$F\text{-measure (F)} = \frac{1}{\left(\frac{1/r + 1/p}{2}\right)} = \frac{2rp}{r + p} = \frac{2a}{2a + b + c} = \frac{2TP}{2TP + FP + FN}$$

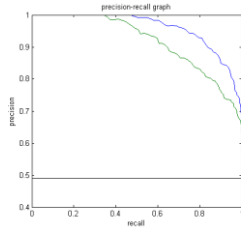
Count	PREDICTED CLASS	
	Class=Yes	Class=No
	Class=Yes	Class=No
Class=Yes	a	b
Class=No	c	d

- Precision is biased towards $C(\text{Yes}|\text{Yes})$ & $C(\text{Yes}|\text{No})$
- Recall is biased towards $C(\text{Yes}|\text{Yes})$ & $C(\text{No}|\text{Yes})$
- F-measure is biased towards all except $C(\text{No}|\text{No})$

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Precision-Recall plot

- Usually for parameterized models, it controls the precision/recall tradeoff



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Sensitivity and Specificity

The sensitivity and specificity measures can be used, respectively, for this purpose. Sensitivity is also referred to as the *true positive (recognition) rate* (i.e., the proportion of positive tuples that are correctly identified), while specificity is the *true negative rate* (i.e., the proportion of negative tuples that are correctly identified). These measures are defined as

$$\text{sensitivity} = \frac{TP}{P}$$

$$\text{specificity} = \frac{TN}{N}$$

It can be shown that accuracy is a function of sensitivity and specificity:

$$\text{accuracy} = \text{sensitivity} \cdot \frac{P}{(P+N)} + \text{specificity} \cdot \frac{N}{(P+N)}.$$

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Model Evaluation

- Metrics for Performance Evaluation
 - How to evaluate the performance of a model?
- Methods for Performance Evaluation
 - How to obtain reliable estimates?



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Methods for Performance Evaluation

- How to obtain a reliable estimate of performance?
- Performance of a model may depend on other factors besides the learning algorithm:
 - Class distribution
 - Cost of misclassification
 - Size of training and test sets

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Methods of Estimation

- Holdout**
 - Reserve 2/3 for training and 1/3 for testing
- Random subsampling**
 - One sample may be biased -- Repeated holdout
- Cross validation**
 - Partition data into k disjoint subsets
 - k -fold: train on $k-1$ partitions, test on the remaining one
 - Leave-one-out: $k=n$
 - Guarantees that each record is used the same number of times for training and testing
- Bootstrap**
 - Sampling with replacement
 - ~63% of records used for training, ~27% for testing

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Cross-validation Example

- 3-fold cross-validation



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Dealing with class Imbalance

- If the class we are interested in is very rare, then the classifier will ignore it.
 - The class imbalance problem
- Solution
 - We can modify the optimization criterion by using a cost sensitive metric
 - We can balance the class distribution
 - Sample from the larger class so that the size of the two classes is the same
 - Replicate the data of the class of interest so that the classes are balanced
 - Over-fitting issues